From Noise to Diversity: Random Embedding Injection in LLM Reasoning

Anonymous Author(s)
Affiliation
Address
`email`

Abstract

Recent *soft prompt* methods, which prepend or insert trainable continuous embeddings into LLM inputs as virtual tokens, optimize these embeddings to enhance mathematical reasoning. However, the mechanism by which embedding injection itself affects model behavior, independent of optimization, remains unexplored. We investigate Random Soft Prompts (RSPs), continuous vectors sampled from the model's token embedding distribution and injected without any training. Empirically, RSPs increase early-stage token entropy while generation stabilizes thereafter, and combining RSPs with temperature sampling yields higher trajectory diversity as measured by Pass@n. Theoretically, we provide two complementary analyses: an energy-based framework showing that RSP introduces Langevin-like perturbations on the Hopfield energy landscape with natural dual decay across layers and generation steps, and a distributional analysis showing that RSP induces stochastic token rankings and expands the output support beyond temperature sampling. We further discuss the implications for GRPO training, where trajectory diversity is critical for learning signal quality.

1 Introduction

As large language models (LLMs) scale to billions of parameters, full fine-tuning becomes increasingly expensive. This has motivated parameter-efficient methods that adapt frozen models by introducing a small number of trainable parameters [Hu et al., 2022, Houlsby et al., 2019]. Among these, *soft prompts* prepend or insert learnable continuous embeddings into the input sequence, steering model behavior through the attention mechanism [Li and Liang, 2021, Lester et al., 2021]. This paradigm has been actively adopted for enhancing mathematical reasoning in LLMs [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Hao et al., 2025, Zhang et al., 2026]. Despite their diverse designs, these methods share a common structure: they inject continuous embeddings that lie outside the pretrained vocabulary and optimize them to improve downstream performance.

However, we find that optimization is not a necessary condition for these effects. We observe that injecting randomly initialized, untrained continuous embeddings can also shift the output distribution. For instance, applying a chat template to a base model not fine-tuned with such format causes a prompt-format mismatch that degrades reasoning capability; appending random embeddings to such mismatched inputs mitigates this degradation. This suggests that injecting out-of-vocabulary continuous embeddings itself, independent of any learned content, plays a non-trivial role in shaping model behavior. Existing studies primarily evaluate *soft prompt* methods through end-task accuracy, and the mechanisms by which embedding injection influences the output distribution remain insufficiently explored.

In this work, we investigate the role of Random Soft Prompts (RSPs), continuous vectors sampled from the Gaussian distribution of the model's token embedding matrix and injected without any training. Empirically, RSPs increase early-stage token entropy while the output stabilizes in later generation stages, and combining RSPs with temperature sampling yields more diverse reasoning trajectories as measured by Pass@n scaling. Theoretically, we analyze RSPs from two perspectives: an energy-based framework showing that RSP introduces controlled stochastic perturbations with natural decay through the attention mechanism, and a distributional analysis showing that RSP induces stochastic token rankings and expands the output support beyond what temperature sampling can achieve. We further discuss the implications of these findings for Group Relative Policy Optimization (GRPO) [Shao et al., 2024], where trajectory diversity is essential for effective learning signals.

Our contributions are as follows:

- We empirically analyze how RSPs affect LLM generation, showing that RSPs increase early token entropy and produce more diverse reasoning trajectories when combined with temperature sampling.
- We provide theoretical analyses from two perspectives: an energy-based framework explaining why RSP can perturb the output distribution without degrading performance, and a distributional analysis showing that RSP induces stochastic token rankings and expands the output support beyond the reach of temperature sampling.
- We discuss the implications of RSP-induced diversity for GRPO training and argue that RSP provides exploration gains complementary to existing temperature-based approaches.

2 Background

2.1 Soft Prompt Methods for LLM Reasoning

Soft prompts emerged as a parameter-efficient alternative to full fine-tuning. Prefix-Tuning [Li and Liang, 2021] introduced the paradigm by prepending trainable continuous vectors to every transformer layer, Prompt Tuning [Lester et al., 2021] simplified this to input-layer-only tuning and showed that it matches full fine-tuning at sufficient scale, and P-tuning [Liu et al., 2024] extended the approach with an LSTM-based prompt encoder to model inter-position dependencies.

More recently, *soft prompts* have been actively applied to LLM reasoning. TTSV [Kang et al., 2026] optimizes prefix embeddings at test time via entropy minimization, steering the model toward high-certainty states to activate latent reasoning capabilities. SoftCoT [Xu et al., 2025] replaces explicit chain-of-thought tokens with soft tokens generated by an assistant model, leveraging the stronger representational capacity of continuous representations over discrete language. LTPO [Ye et al., 2026] optimizes latent thought embeddings per test instance through policy gradient with a confidence-based intrinsic reward. COCONUT [Hao et al., 2025] feeds the model's own hidden states back as input embeddings, enabling reasoning in a continuous latent space. MemGen [Zhang et al., 2026] demonstrates that embedding vectors can serve as experience memory by constructing latent token sequences that enrich the model's reasoning. Pause tokens [Goyal et al., 2024] insert learnable tokens into the input to provide additional computation steps.

These approaches all inject out-of-vocabulary continuous embeddings optimized for task-specific objectives. Because evaluation centers on end-task accuracy, the contribution of the injection itself and that of the optimization are not separated, and the mechanisms by which such out-of-vocabulary continuous embeddings shape model behavior remain a separate and underexplored question.

2.2 Noise Injection in Neural Networks

Neural networks have incorporated noise at various stages. During training, dropout [Srivastava et al., 2014] randomly deactivates hidden units to prevent overfitting, and NEFTune [Jain et al., 2024] adds noise to token embeddings during instruction fine-tuning, reporting improvements in instruction following. These methods use noise as a regularizer and do not apply perturbations at inference time.

At inference time, perturbation-based approaches primarily target robustness and safety. Randomized smoothing [Cohen et al., 2019] injects Gaussian noise into inputs to obtain certified robustness guarantees. In the LLM setting, SmoothLLM [Robey et al., 2025] perturbs input prompts at the character level to defend against adversarial jailbreaking. These methods require multiple noisy forward passes with output aggregation, and target prediction stability rather than output diversity.

Table 1: Accuracy (%) across model configurations and benchmarks. RSP reports the best result across injection positions (Prefix, Infix, Suffix); full per-position breakdown is in Appendix A. Values in parentheses indicate the difference from the baseline.

	MATH-500		GSM8K		AIME24	
Model	Baseline	RSP	Baseline	RSP	Baseline	RSP
Instruct Models						
LLaMA-3.1-8B-Inst	52.20	49.40 (−2.8)	85.75	86.73 (+1.0)	6.67	10.00 (+3.3)
Qwen2.5-Math-7B-Inst	83.20	84.20 (+1.0)	95.45	95.68 (+0.2)	13.33	16.67 (+3.3)
Qwen2.5-Math-1.5B-Inst	73.00	75.20 (+2.2)	85.29	85.75 (+0.5)	10.00	20.00 (+10.0)
Base Models (Raw Text)						
Qwen2.5-Math-7B	72.20	71.60 (−0.6)	84.15	84.31 (+0.2)	6.67	16.67 (+10.0)
Qwen2.5-Math-1.5B	61.40	64.40 (+3.0)	77.86	74.30 (−3.6)	16.67	10.00 (−6.7)
Base Models + ChatML (Format Mismatch)						
Qwen2.5-Math-7B	52.20	70.40 (+18.2)	58.30	75.97 (+17.7)	23.33	23.33 (+0.0)
Qwen2.5-Math-1.5B	34.40	63.40 (+29.0)	37.45	67.02 (+29.6)	13.33	20.00 (+6.7)

In reinforcement learning, NoisyNet [Fortunato et al., 2018] injects learnable noise into network weights to improve exploration, demonstrating that the location and structure of noise injection affect exploration quality. This motivation is closest to RSPs, which similarly target diversity, but differ in injection target (input embeddings vs. network weights) and require no learned noise parameters.

RSPs differ from these approaches in the following respects. **(1)** RSPs operate solely at inference without any training, whereas dropout and NEFTune inject noise during training, modifying the resulting model parameters accordingly. **(2)** RSPs preserve the original input and append fresh embedding vectors in a single forward pass, whereas robustness methods corrupt the input and aggregate over multiple noisy passes. **(3)** RSPs use purely random vectors sampled from the pretrained embedding statistics, whereas NoisyNet learns noise parameters through optimization.

3 Experimental Analysis

3.1 Random Soft Prompt: Definition and Setup

Let $\mathbf{W}_E \in \mathbb{R}^{V \times d}$ denote the token embedding matrix of a pretrained LLM, where V is the vocabulary size and d is the hidden dimension. We define an RSP $\mathbf{H} = [h_1, \ldots, h_L] \in \mathbb{R}^{L \times d}$ as a sequence of L continuous vectors, where each vector is independently sampled from:

$$h_j \sim \mathcal{N}(\mu_E,\ \sigma_E^2 \mathbf{I}), \quad j = 1, \ldots, L, \tag{1}$$

with $\mu_E = \frac{1}{Vd} \sum_{i,k} [\mathbf{W}_E]_{i,k}$ and $\sigma_E = \mathrm{std}(\mathbf{W}_E)$. Let $\mathbf{X} = [x_1, \ldots, x_T] \in \mathbb{R}^{T \times d}$ denote the embedding sequence of the original input. The RSP-augmented input $\tilde{\mathbf{X}}$ is constructed by concatenating $\mathbf{H}$ with $\mathbf{X}$ at one of three positions without modifying the original prompt text: *prefix* ($[\mathbf{H}; \mathbf{X}]$), *infix* ($\mathbf{H}$ inserted within $\mathbf{X}$), or *suffix* ($[\mathbf{X}; \mathbf{H}]$), following the injection strategies adopted in prior work [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Zhang et al., 2026]. Sampling a new $\mathbf{H}$ for each batch ensures that the results are not dependent on any realization of the random embeddings.

We evaluate on three mathematical reasoning benchmarks: MATH-500 [Lightman et al., 2024], GSM8K [Cobbe et al., 2021], and AIME24, using LLaMA-3.1-8B-Instruct [Grattafiori et al., 2024] and the Qwen2.5-Math model family [Yang et al., 2024] across seven configurations including instruct models, base models, and base models with mismatched chat templates. All experiments use greedy decoding to isolate the effect of RSPs from sampling stochasticity. Only the final answer is evaluated through a fallback-based extraction pipeline. Full experimental details are provided in Appendix A.

3.2 Does RSP Degrade Reasoning Performance?

We first examine how injecting random embeddings affects the model's reasoning performance. Table 1 reports the accuracy for each model configuration, using the best RSP result across injection positions (Prefix, Infix, Suffix) and $L \in \{10, 15, 20\}$ for each benchmark.

Table 2: Comparison with existing soft prompt methods (TTSV, SoftCoT, LTPO) under unified evaluation. Baseline and RSP values are from Table 1. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Benchmark	Baseline	RSP	TTSV	SoftCoT	LTPO
Qwen2.5-Math-7B-Instruct	MATH-500	83.20	**84.20**	83.80	82.80	83.00
	GSM8K	95.45	**95.68**	95.30	95.00	94.84
	AIME24	13.33	16.67	**23.30**	16.67	13.33
Qwen2.5-Math-1.5B-Instruct	MATH-500	73.00	75.20	75.20	**76.00**	72.40
	GSM8K	85.29	**85.75**	85.70	84.46	84.53
	AIME24	10.00	**20.00**	6.70	6.67	10.00

For instruct models and base models with their native prompt formats, RSP injection maintains or slightly shifts baseline performance within a few percentage points. Most strikingly, under prompt-format mismatch — where base models receive ChatML templates they were not trained on — RSP yields pronounced improvements of up to $+29\%$p. These gains are difficult to attribute to simple perturbation effects, as noise that merely raises model uncertainty would further degrade performance.

We compare RSPs with prior soft prompt methods (TTSV, SoftCoT, LTPO) that optimize their embeddings for distinct objectives, under a unified evaluation protocol reported in Table 2. All methods are re-evaluated with a shared answer extraction pipeline that applies fallback strategies, enabling a format-agnostic comparison despite each method's distinct prompt format and grading scheme. The results place RSPs on par with the optimized methods without requiring any training.

3.3 How Does RSP Affect the Output Distribution?

We analyze how RSP injection changes the model's output distribution during generation on Qwen2.5-Math-1.5B-Instruct with MATH-500, measuring per-token entropy, top-1 probability, and varentropy across the generation process. Figure 1 and Table 3 compare these metrics for the first 5% of generation steps across RSP variants and prior soft prompt methods.

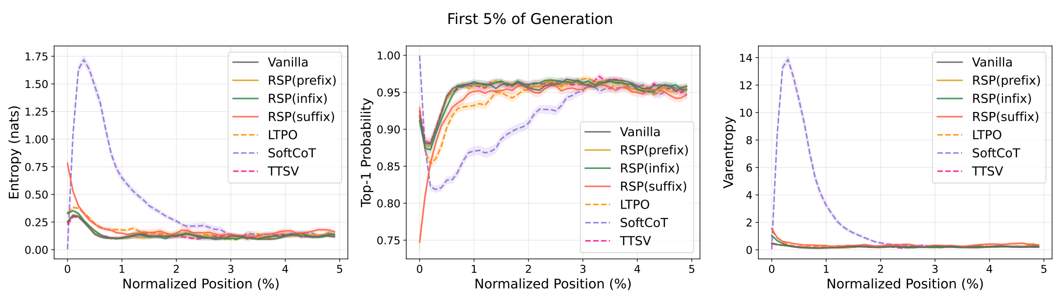

Figure 1: Mean entropy, top-1 probability, and varentropy during the first 5% of generation steps (Qwen2.5-Math-1.5B-Instruct, MATH-500). Shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: existing soft prompt methods (LTPO, SoftCoT, TTSV).

Table 3: Mean entropy, top-1 probability, and varentropy for the first 5% of generation steps (Qwen2.5-Math-1.5B-Instruct, MATH-500).

Metric	Baseline	Prefix	Infix	Suffix	LTPO	SoftCoT	TTSV
Entropy	0.1332	0.1366	0.1402	0.1941	0.1662	0.4218	0.1359
Top-1 Prob	0.9535	0.9523	0.9525	0.9373	0.9437	0.9156	0.9534
Varentropy	0.2181	0.2207	0.2463	0.4010	0.2953	2.2234	0.2174

These results show that RSP modifies the early-stage output distribution across all injection positions: entropy increases (the token distribution flattens), top-1 probability decreases (commitment to a dominant token weakens), and varentropy increases (indicating multimodal distributions [Ahmed et al., 2026]). High-entropy positions are known to act as critical forking points that determine reasoning trajectories [Wang et al., 2025], so this shift moves the model toward a state where multiple reasoning paths can coexist. The change is localized to the initial reasoning stage; in the middle and final portions of generation, all three metrics converge to baseline levels.

Although existing soft prompt methods are optimized for different objectives, they share a common pattern of increasing early-stage entropy. Notably, TTSV, which optimizes soft prompts using a reward that reduces the mean entropy of reasoning trajectories, still exhibits early entropy comparable to the baseline. This suggests that elevating early-stage entropy is an inherent effect of injecting non-pretrained continuous embeddings, independent of the optimization objective.

3.4 Does RSP Induce Trajectory Diversity?

Section 3.3 empirically established that RSP reshapes the early-stage output distribution. We now examine whether this shift translates into reasoning diversity at three complementary levels of analysis: token rankings, semantic content of trajectories, and task-level outcomes. All three analyses share the setting: Qwen2.5-Math-1.5B-Instruct, MATH-500, suffix injection, and $L = 10$.

Token-level: distribution beyond temperature. Under autoregressive decoding, any trajectory-level divergence between RSP and the baseline must originate from the first generation step, a critical forking point for reasoning trajectories [Wang et al., 2025]. Using the baseline at τ=1.0 as reference, we compare RSP at τ=1.0 against the baseline at τ=2.0 (the high-temperature ceiling of what flattening alone can achieve) on three complementary metrics: Spearman rank correlation ρ to detect rank reorganization, the probability mass placed outside the baseline's top-10 to measure support expansion, and Jensen–Shannon divergence to quantify the overall magnitude of distributional change (formal definitions in Appendix C).

Table 4: First-token distribution metrics measured against the baseline at τ=1.0, comparing the baseline at τ=2.0 with RSP at τ=1.0.

Metric	τ=2.0	RSP
Spearman ρ	1.0	**0.651**
Mass outside top-10	5.15%	**27.64%**
JS divergence	0.086	**0.231**

Table 4 reports the contrast. Even at τ=2.0, only 5.15% of probability mass falls outside the baseline's top-10; RSP at τ=1.0 reaches 27.64%. The Spearman rank correlation tells the story more starkly: because temperature scaling $p_i^{(\tau)} \propto p_i^{1/\tau}$ is a monotone transform that preserves token ranking ($p_i > p_j \iff p_i^{(\tau)} > p_j^{(\tau)}$) for any $\tau > 0$, ρ relative to the baseline is exactly 1 as a mathematical identity. RSP, by contrast, reranks the token distribution itself, reducing ρ to 0.651—an effect temperature cannot produce at any τ. The Jensen–Shannon divergence (0.231) quantifies the overall magnitude of this distributional change. Since $\rho < 1$ rules out rank-preserving flattening as the sole cause, the high divergence reflects a structural reshape of the distribution rather than a uniform flattening of the same ranking.

Trajectory-level: semantic diversity. Token-level reranking raises a second question: do first-token perturbations produce semantically distinct reasoning paths, or do independently perturbed trajectories converge to the same solution strategy? We randomly sample 64 problems from MATH-500 and generate 300 independent trajectories per problem under two conditions, Baseline and RSP, with temperature $\tau = 1.0$. The trajectories are embedded via BGE-M3 [Chen et al., 2024] and compared on three metrics: mean pairwise cosine distance within each problem, inter-cluster distance between DBSCAN [Ester et al., 1996] centroids, and intra-cluster distance. Table 5 shows that RSP substantially increases the inter-cluster distance while leaving the intra-cluster distance unchanged: clusters become more separated while reasoning within each cluster remains consistent. RSP also wins on pairwise distance in 56 of 64 problems (87.5%).

Table 5: Semantic diversity metrics averaged across the 64 problems.

Metric	Baseline	RSP
Pairwise cos. dist.	0.0612	**0.0879**
Inter-cluster dist.	0.0183	**0.0982**
Intra-cluster dist.	0.0526	0.0528

Figure 2(a) illustrates the qualitative difference for the representative problem: the RSP samples (blue) cover the region occupied by baseline samples (red) while extending into clusters that the baseline does not reach; the expanded regions also contain correct solutions (filled markers), showing that RSP's exploration is not limited to incorrect alternatives but reaches new reasoning paths that still yield the correct answer. Figures 2(b) and 2(c) show that this pattern holds across the 64 problems: on most problems RSP produces more clusters and larger pairwise cosine distance than baseline. The combination of large inter-cluster increase and unchanged intra-cluster distance mirrors the token-

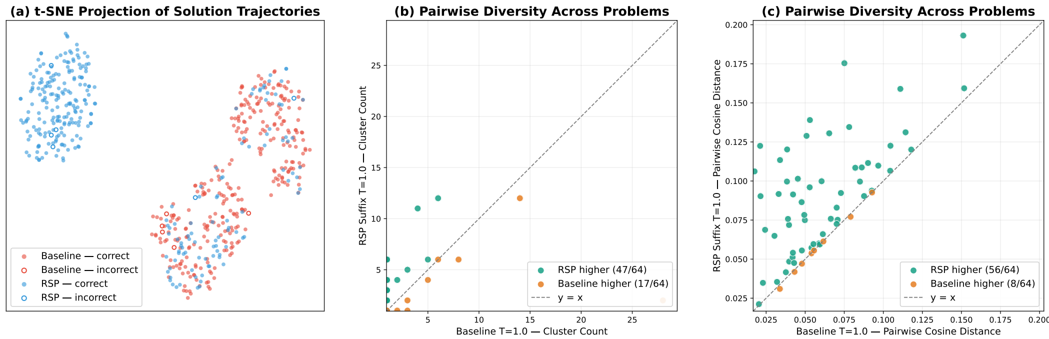

Figure 2: Semantic diversity analysis on 64 randomly sampled MATH-500 problems with 300 trajectories per condition (Qwen2.5-Math-1.5B-Instruct, $\tau = 1.0$, BGE-M3 embeddings). (a) t-SNE projection for one representative problem (`test/prealgebra/951`): baseline (red) concentrates around one region while RSP suffix (blue) spreads across distinct clusters; filled markers denote correct answers and open markers denote incorrect. (b) Per-problem mean pairwise cosine distance; points above the diagonal indicate RSP produces more diverse trajectories than baseline. (c) Per-problem DBSCAN cluster count; RSP yields more distinct clusters than baseline on most problems.

level finding: RSP alters which cluster a trajectory enters while preserving the local continuation within each cluster. Additional per-problem t-SNE plots are provided in Figure 6 (Appendix B).

Outcome-level: Pass@n scaling. Finally, we evaluate whether the token- and trajectory-level diversity translates into task-level outcome diversity through Pass@n, the probability that at least one of n sampled solutions is correct. We compare four conditions with $n = 16$ samples per problem on MATH-500 and AIME24: *(i) Baseline,* $\tau = 1.0$: temperature sampling only; *(ii) RSP (fixed seed),* $\tau = 1.0$: a single RSP shared across all samples; *(iii) RSP (indep. seed), greedy*: a different RSP per sample without temperature; and *(iv) RSP (indep. seed),* $\tau = 1.0$: a different RSP per sample combined with temperature sampling.

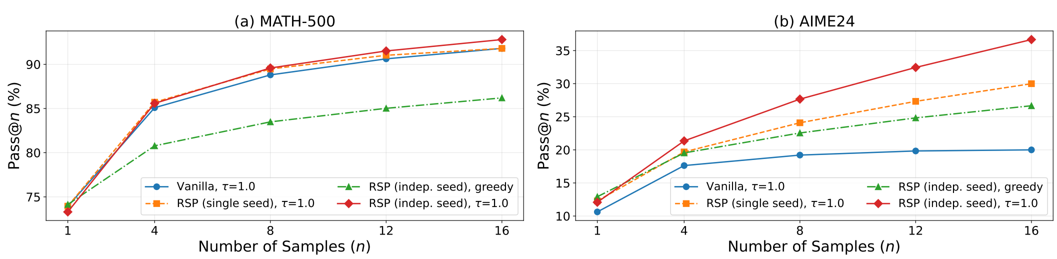

Figure 3: Pass@n scaling on MATH-500 (left) and AIME24 (right) with Qwen2.5-Math-1.5B-Instruct, $n = 16$ samples per problem. Baseline: temperature sampling only. Fixed seed: single RSP shared across samples combined with temperature. Indep. seed: a different RSP per sample, with or without temperature.

Condition *(iv)* achieves the highest Pass@n on both benchmarks, reaching 92.80% on MATH-500 and 36.67% on AIME24 at $n = 16$. Condition *(ii)* shows no improvement over the baseline, confirming that the diversity gain originates from varying the random embeddings across samples, not from a single fixed perturbation. Pass@1 remains comparable to the baseline on MATH-500, and on the more challenging AIME24, *(iv)* achieves higher Pass@1 than the baseline. Independent RSPs combined with temperature sampling produce the most diverse reasoning trajectories while maintaining or improving single-sample accuracy.

Synthesis across the three levels. The three analyses converge on a consistent picture. At the token level, RSP induces support expansion concentrated at the first token and decaying monotonically across steps. At the trajectory level, this localized perturbation translates into distinct reasoning strategies (large inter-cluster separation) while preserving local coherence within each strategy (unchanged intra-cluster distance). At the outcome level, the resulting trajectories cover more correct solutions without sacrificing single-sample accuracy.

4 Theoretical Analysis: Why RSP Works

4.1 RSP as Controlled Exploration on the Energy Landscape

Under the two simplifying assumptions introduced below, this section provides a theoretical account of two properties observed in our experiments: (1) RSP can change the model's output distribution, and (2) RSP does not degrade reasoning performance. We show that RSP converts the transformer's deterministic forward pass into a controlled stochastic process on the Hopfield energy landscape, where the noise introduced through attention naturally decays with both layer depth and generation step. We build on the equivalence between softmax attention and gradient descent on the Modern Hopfield energy [Ramsauer et al., 2021]. The analysis relies on two simplifying assumptions: (A1) $\mathbf{K} = \mathbf{V}$ in attention (exact in Hopfield models, approximate in standard transformers) and (A2) weight-tied layers (exact in deep equilibrium models [Bai et al., 2019], approximate otherwise). Full proofs are in Appendix D.

Transformer layers as energy minimization. The Modern Hopfield Energy for stored patterns $\boldsymbol{\Xi} \in \mathbb{R}^{d\times m}$ and state $\boldsymbol{\sigma} \in \mathbb{R}^d$ is $E(\boldsymbol{\sigma}; \boldsymbol{\Xi}) = -\beta^{-1} \log \sum_{j=1}^{m} \exp(\beta\, \boldsymbol{\xi}_j^\top \boldsymbol{\sigma}) + \frac{1}{2}\|\boldsymbol{\sigma}\|^2$, where $\beta = 1/\sqrt{d}$. A full gradient step on this energy yields $\boldsymbol{\sigma}^{\mathrm{new}} = \boldsymbol{\Xi} \cdot \mathrm{softmax}(\beta\, \boldsymbol{\Xi}^\top \boldsymbol{\sigma})$, which is identical to softmax attention [Ramsauer et al., 2021]. Each transformer layer thus performs one gradient descent step on this energy landscape: the hidden state moves toward a lower-energy configuration at every layer.

The trapping problem. This monotone descent implies that, without perturbation, the forward pass converges to the local minimum in the basin of attraction of $\boldsymbol{\sigma}^{(0)}$. If a better minimum $\boldsymbol{\sigma}_g^*$ exists in a different basin, the deterministic forward pass cannot reach it because it cannot cross energy barriers.

Why temperature scaling and additive noise do not resolve this. Temperature scaling modifies only the final output distribution: $p^{(\tau)}(v) = \mathrm{softmax}(\mathbf{z}^{(L)}/\tau)$, leaving the internal trajectory $\mathbf{x}^{(l+1)}|_\tau = \mathbf{x}^{(l+1)}|_{\tau=1}$ for all $l < L$ identical. It cannot help escape a local minimum during inference. Additive Gaussian noise $\mathbf{x}^{(0)} + \boldsymbol{\epsilon}$ perturbs the initial state, but in subsequent layers the perturbation magnitude is bounded by $\prod_{l'} L_{f_{l'}} \cdot \sigma$ where $L_{f_{l'}}$ is the Lipschitz constant of layer l'. When $L_{f_{l'}} > 1$, the noise amplifies rather than decays, providing no controlled exploration schedule.

RSP as controlled exploration via attention. RSP resolves both limitations through two mechanisms.

First, concatenating k random vectors sampled from $\mathcal{N}(\mu_E, \sigma_E^2 \mathbf{I})$ causes each token to attend to both real and random keys, yielding

$$\mathbf{x}_i^{(l+1)} = (1 - w_{r,i}^{(l)})\, \tilde{\mathbf{x}}_i^{(l+1)} + \boldsymbol{\eta}_i^{(l)}, \tag{2}$$

where $\tilde{\mathbf{x}}_i^{(l+1)}$ is the standard attention update over real tokens, and $\boldsymbol{\eta}_i^{(l)} := \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)}$ aggregates the contribution of random tokens with total attention mass $w_{r,i}^{(l)} := \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)}$.

The perturbation magnitude is controlled by $w_{r,i}^{(l)}$, with variance scaling on the order of $(w_{r,i}^{(l)})^2$. In practice, since the embedding mean is negligible relative to its standard deviation ($|\mu_E|/\sigma_E \ll 1$), the bias term is dominated by the stochastic component, and the dynamics are well-approximated by a gradient descent process with approximately zero-mean structured noise (see Appendix D.2).

This yields a Langevin-like stochastic process that enables exploration beyond local minima.

Second, RSP modifies the energy landscape. Adding k random keys changes the Hopfield energy as

$$\tilde{E}_i - E_i = -\beta^{-1} \log\left(1 + \frac{B_i}{A_i}\right) \leq 0, \tag{3}$$

where $A_i = \sum_j \exp(\beta\, \mathbf{k}_j^\top \mathbf{q}_i)$ and $B_i = \sum_j \exp(\beta\, \mathbf{k}_{r_j}^\top \mathbf{q}_i) > 0$. The magnitude scales with $w_{r,i}^{(l)}/(1 - w_{r,i}^{(l)})$, so tokens assigning more attention to random keys experience larger energy perturbations.

While this does not directly reduce barrier height, it alters the local energy landscape, and together with stochastic perturbations increases the probability of crossing energy barriers between basins.

Both mechanisms are governed by $w_{r,i}^{(l)}$, which controls the overall strength of exploration.

Why performance is preserved: dual decay of $w_{r,i}^{(l)}$. The attention decay bound is:

$$w_{r,i}^{(l)} \leq \frac{k}{k + n \cdot \exp(\Delta_i^{(l)}/\sqrt{d})}, \tag{4}$$

where $\Delta_i^{(l)} := \min_{j\in[n]} \mathbf{q}_i^\top \mathbf{k}_j - \max_{j'\in[k]} \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}$ is the real-vs-random logit gap and n is the number of real tokens. Since the bias, stochastic scale, and energy modification are all controlled by $w_{r,i}^{(l)}$, this bound governs the overall perturbation strength.

Layer-wise decay. If deeper layers increase the real-vs-random logit gap $\Delta_i^{(l)}$, then $w_{r,i}^{(l)}$ correspondingly decays toward zero. Empirically, as deeper layers develop task-specific representations, $\Delta_i^{(l)}$ tends to increase, making RSP's perturbation strongest in early layers and negligible in deeper layers.

Sequence-wise decay. In autoregressive generation, each newly generated token adds a real key to the KV cache, so n increases at every generation step while k remains fixed. The growing n in Eq. (4) causes $w_{r,i}^{(l)}$ to decrease monotonically, progressively diluting RSP's influence as the sequence grows.

This dual decay provides a natural annealing schedule: RSP enables exploration in the early stage of both the internal forward pass and the generation sequence, while the direct influence of the perturbation progressively diminishes as the model accumulates real tokens and converges to task-specific representations.

Empirical verification. We empirically verify this dual decay on Qwen2.5-Math-7B with suffix injection over 500 MATH-500 problems. For each problem, we measure the per-token attention mass directed to the RSP embeddings and to the question tokens within the same sequence, with the question tokens serving as a baseline that factors out sequence-length effects. Figure 4 shows the expected contrast: the question tokens, which remain task-relevant throughout reasoning, receive attention whose distribution varies across layers and reasoning positions without a consistent trend, while the RSP embeddings, which carry no task-relevant content and act as noise, receive attention that decreases along both the layer and sequence axes, directly instantiating the layer-wise and sequence-wise dilution predicted by Eq. (4).

4.2 Output Distribution Shift and Exploration

We study how Random Soft Prompting (RSP) alters the output distribution of an autoregressive language model and how this leads to improved exploration.

In autoregressive decoding, generation proceeds via prefix-conditioned distributions $p(y_t \mid x, y_{<t})$. RSP does not modify a single global distribution, but perturbs the sequence of conditional distributions across decoding steps. In Section 4.1, we analyzed this from an internal, layer-wise and sequence-wise perspective, showing that RSP introduces structured perturbations through attention that naturally decay across layers. We now examine how these perturbations manifest at the output level.

Limitation of temperature scaling. Temperature scaling transforms probabilities as $p_i^{(\tau)} \propto p_i^{1/\tau}$ and preserves token ranking, i.e., $p_i > p_j \iff p_i^{(\tau)} > p_j^{(\tau)}$. Hence it cannot introduce new token orderings at a fixed decoding state.

Effect of RSP. RSP induces stochastic perturbations to the hidden state, which we locally approximate as $z_i(h) \approx z_i + a_i^\top h$, where h is a random soft prompt.

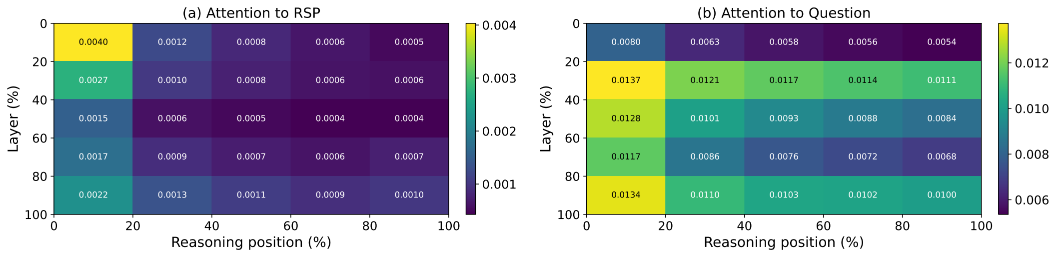

Figure 4: Per-token attention mass on Qwen2.5-Math-7B (suffix, 500 MATH-500 problems, each with an independently sampled RSP). The layer and reasoning-position axes are each partitioned into five quantile bins; every cell reports the 500-sample mean per-token attention mass directed to **(a)** the RSP embeddings and **(b)** the question tokens within that region. The horizontal axis spans reasoning position from early (left) to late (right) generation steps, and the vertical axis spans layer depth from shallow (top) to deep (bottom). Attention to the RSP embeddings in (a) decays along both axes — concentrating at shallow layers and early reasoning positions — while attention to the question tokens in (b) remains comparatively uniform, visually confirming the layer-wise and sequence-wise dilution predicted by Eq. (4).

Proposition 1 (Stochastic ranking). If there exists a token pair (i, j) such that $a_i \neq a_j$, then

$$0 < \Pr_h \big(z_i(h) > z_j(h)\big) < 1.$$

Thus, unlike temperature scaling, RSP randomizes token rankings at each decoding step.

Proposition 2 (Support expansion). Let $\mathrm{TopK}_{\mathrm{Temp}}(s_t)$ be the top-K tokens under temperature scaling at state s_t. Under the same condition, there exists $i \notin \mathrm{TopK}_{\mathrm{Temp}}(s_t)$ such that

$$\Pr_h \big(i \in \mathrm{TopK}(p^{\mathrm{RSP}}(\cdot \mid s_t, h))\big) > 0.$$

Implication. RSP expands the effective support of the output distribution, enabling tokens that are effectively inaccessible under temperature scaling. Since autoregressive generation composes these distributions over time, such perturbations can lead to distinct trajectories, explaining the empirical gains in exploration.

5 Conclusion

Summary. This work investigated the role of Random Soft Prompts (RSPs), continuous embeddings sampled from the pretrained token embedding distribution and injected into the model input without any training. Through empirical and theoretical analyses, we established three findings. First, RSPs increase early-stage token entropy, top-1 probability reduction, and varentropy while the model's output stabilizes in later generation stages (Section 3). Second, RSPs induce stochastic token rankings and expand the effective support of the output distribution, a property that temperature sampling fundamentally lacks due to its ranking-preserving nature (Section 4). Third, combining RSPs with temperature sampling produces more diverse reasoning trajectories than either method alone, as measured by Pass@n scaling (Section 3).

Implications for GRPO training. GRPO computes advantages by comparing multiple sampled trajectories within each group, making trajectory diversity essential for effective learning signals. Existing approaches such as DAPO [Yu et al., 2025] and CE-GPPO [Su et al., 2026] address entropy collapse by modifying the clipping range or reintroducing gradients from clipped tokens, but the underlying sampling process remains based on temperature sampling, which preserves token rankings and cannot expand the output support. Our theoretical and empirical results suggest that RSP can address this limitation at the sampling level: by perturbing the initial output distribution with independent random embeddings per rollout, RSP provides a complementary source of diversity that is orthogonal to temperature-based exploration. We expect this to yield more effective learning signals for GRPO training, particularly in mitigating entropy collapse.

Limitations. Our experiments are conducted on mathematical reasoning benchmarks (MATH-500, GSM8K, AIME24) with a limited set of model families (Qwen2.5-Math, LLaMA-3.1). The energy-theoretic analysis in Section 4 relies on strong assumptions: the Hopfield–Attention equivalence (A1: $\mathbf{K} = \mathbf{V}$) and weight-tied layers (A2), which hold exactly only in Hopfield-type models and deep equilibrium models, respectively. Since real transformer architectures vary widely in their attention mechanisms and layer structures, the theoretical predictions are most directly applicable to models whose attention dynamics closely approximate the Hopfield energy framework. The GRPO application remains a theoretical expectation and has not been empirically validated.

Future work. Extending the experimental evaluation to other domains (e.g., code generation, commonsense reasoning) and larger model scales would test the generality of RSP's effects. Empirically validating the GRPO application by training with RSP-augmented rollouts and comparing against DAPO and CE-GPPO is a natural next step. Investigating the optimal RSP distribution and token count, which currently require per-model tuning, could lead to more principled injection strategies.

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A Experimental Setup and Full Accuracy Breakdown

Table 6 provides the full per-position accuracy breakdown (Prefix, Infix, Suffix) corresponding to the best-across-positions results summarized in Table 1 in the main text. Table 7 lists the number of RSP tokens L used for each model–benchmark pair, selected from $\{10, 15, 20\}$. All experiments are conducted on a single NVIDIA A6000 40GB GPU with greedy decoding, a maximum generation length of 3,072 tokens for MATH-500 and GSM8K, and 4,096 tokens for AIME24. Batch size is fixed per model (16 for LLaMA-3.1-8B-Instruct and Qwen2.5-Math-7B variants, 32 for Qwen2.5-Math-1.5B variants).

Table 6: Accuracy (%) across model configurations, injection positions, and benchmarks. Values in parentheses indicate the difference from the baseline. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Mode	MATH-500	GSM8K	AIME24
Instruct Models				
LLaMA-3.1-8B-Instruct	Baseline	**52.20**	85.75	6.67
	Prefix	45.00 (−7.2)	76.35 (−9.4)	3.33 (−3.3)
	Infix	49.40 (−2.8)	85.97 (+0.2)	**10.00** (+3.3)
	Suffix	49.40 (−2.8)	**86.73** (+1.0)	**10.00** (+3.3)
Qwen2.5-Math-7B-Instruct	Baseline	83.20	95.45	13.33
	Prefix	81.40 (−1.8)	94.92 (−0.5)	13.33 (+0.0)
	Infix	**84.20** (+1.0)	95.60 (+0.2)	**16.67** (+3.3)
	Suffix	83.40 (+0.2)	**95.68** (+0.2)	**16.67** (+3.3)
Qwen2.5-Math-1.5B-Instruct	Baseline	73.00	85.29	10.00
	Prefix	74.80 (+1.8)	85.29 (+0.0)	13.33 (+3.3)
	Infix	**75.20** (+2.2)	**85.75** (+0.5)	6.67 (−3.3)
	Suffix	74.20 (+1.2)	84.69 (−0.6)	**20.00** (+10.0)
Base Models (Raw Text)				
Qwen2.5-Math-7B	Baseline	**72.20**	84.15	6.67
	Prefix	71.60 (−0.6)	**84.31** (+0.2)	**16.67** (+10.0)
	Infix	63.20 (−9.0)	64.29 (−19.9)	**16.67** (+10.0)
	Suffix	55.60 (−16.6)	54.13 (−30.0)	13.33 (+6.7)
Qwen2.5-Math-1.5B	Baseline	61.40	**77.86**	**16.67**
	Prefix	**64.40** (+3.0)	74.30 (−3.6)	10.00 (−6.7)
	Infix	63.20 (+1.8)	73.92 (−3.9)	10.00 (−6.7)
	Suffix	54.60 (−6.8)	58.61 (−19.3)	3.33 (−13.3)
Base Models + ChatML (Format Mismatch)				
Qwen2.5-Math-7B	Baseline	52.20	58.30	**23.33**
	Prefix	59.20 (+7.0)	56.41 (−1.9)	3.33 (−20.0)
	Infix	62.00 (+9.8)	69.07 (+10.8)	**23.33** (+0.0)
	Suffix	**70.40** (+18.2)	**75.97** (+17.7)	**23.33** (+0.0)
Qwen2.5-Math-1.5B	Baseline	34.40	37.45	13.33
	Prefix	**63.40** (+29.0)	**67.02** (+29.6)	**20.00** (+6.7)
	Infix	39.20 (+4.8)	50.42 (+13.0)	6.67 (−6.7)
	Suffix	51.00 (+16.6)	50.64 (+13.2)	13.33 (+0.0)

Table 7: Number of RSP tokens (L) used for each model and benchmark.

Model	MATH-500	GSM8K	AIME24
LLaMA-3.1-8B-Instruct	15	20	15
Qwen2.5-Math-7B-Instruct	20	20	20
Qwen2.5-Math-1.5B-Instruct	20	10	20
Qwen2.5-Math-7B	10	10	20
Qwen2.5-Math-1.5B	10	10	20
Qwen2.5-Math-1.5B (ChatML)	20	20	10
Qwen2.5-Math-7B (ChatML)	20	20	20

A.1 RSP Injection Positions and Prompt Templates

Below we show the prompt structure for each injection position across all model types. Blue text indicates RSP embeddings, and purple text indicates special tokens.

A.1.1 Prefix

RSP embeddings are concatenated before the prompt embeddings. No text modification is applied.

ChatML (Qwen Instruct / Base + ChatML).

```
[Random Embeddings (L tokens)]
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
[Random Embeddings (L tokens)]
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date:  December 2023
Today Date:  26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
[Random Embeddings (L tokens)]
{question}
Please reason step by step, and put your final answer within \boxed{}.
```

A.1.2 Infix

A description text and special tokens are inserted into the prompt. The special token positions are then replaced with RSP embeddings at the embedding level.

ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning.  If these tokens are useful for your case, you can
use them as reference.  If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens:  <special_token_1>...<special_token_L>
<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date:  December 2023
Today Date:  26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
```

```
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens:
<reserved_special_token_0>...
<reserved_special_token_L>
<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens: <special_token_1>...<special_token_L>
Please reason step by step, and put your final answer within \boxed{}.
```

A.1.3 Suffix

For chat-templated models, RSP embeddings are inserted immediately before the end-of-turn token. For raw text models, RSP embeddings are concatenated at the end of the prompt.

ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}[Random Embeddings (L tokens)]<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}[Random Embeddings (L tokens)]<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
{question}
Please reason step by step, and put your final answer within \boxed{}.[Random
Embeddings (L tokens)]
```

A.2 Answer Extraction and Grading

The final answer is extracted from the model output through a fallback chain. Each step is attempted in order; if extraction fails, the next step is tried.

Answer Extraction Fallback Chain

1. `"final answer is $...$. I hope"` pattern (Minerva format)
2. `\boxed{...}`: last non-empty box is used; empty `\boxed{}` are skipped
3. Text following `"the answer is"`
4. Text following `"final answer is"`
5. Last number in the output (regex fallback)

Extracted answers are normalized by removing `$`, `\left`, `\right`, and unit strings. Grading is performed via symbolic equivalence checking: exact string match, numerical comparison (`math.isclose`, `rel_tol=1e-4`), and SymPy symbolic simplification.

A.3 Reproduced Prior Work Hyperparameters

All prior methods are reproduced on Qwen2.5-Math-1.5B-Instruct and Qwen2.5-Math-7B-Instruct, evaluated with greedy decoding (temperature $= 0$) and our unified answer extraction pipeline.

TTSV. Prefix length 20, trained for 20 epochs with AdamW optimizer, batch size 2, gradient accumulation 8 steps, and linear learning rate schedule. Learning rate is 1e-3 for Qwen-1.5B and 1e-5 for Qwen-7B.

LTPO. Table 8 reports the per-benchmark hyperparameters.

Table 8: LTPO hyperparameters.

Parameter	GSM8K	MATH-500	AIME24
tokens	8	8	8
steps	20	20	20
top_k	10	10	10
sigma	20	20	5
sigma_decay	0.95	0.95	0.95
lr	1e-2	5e-3	5e-2

SoftCoT. Projection module trained for 10 epochs. Thought tokens: 32 during training, 4 during evaluation. Batch size 4 for GSM8K, 1 for MATH with gradient accumulation 4. The small (assistant) model is Qwen2.5-Math-1.5B-Instruct in both configurations, while the large model is Qwen2.5-Math-1.5B-Instruct (learning rate 2e-5) or Qwen2.5-Math-7B-Instruct (learning rate 5e-6). For MATH-500 and AIME24 evaluation, we use the projection module trained on GSM8K, as it yields higher accuracy than the one trained on MATH.

A.4 Full Entropy Curves

Figure 5 shows the full normalized (0–100%) generation trajectories for entropy, top-1 probability, and varentropy. All methods converge to similar levels after the initial generation stage, confirming that the distributional changes induced by RSP are localized to the early reasoning phase.

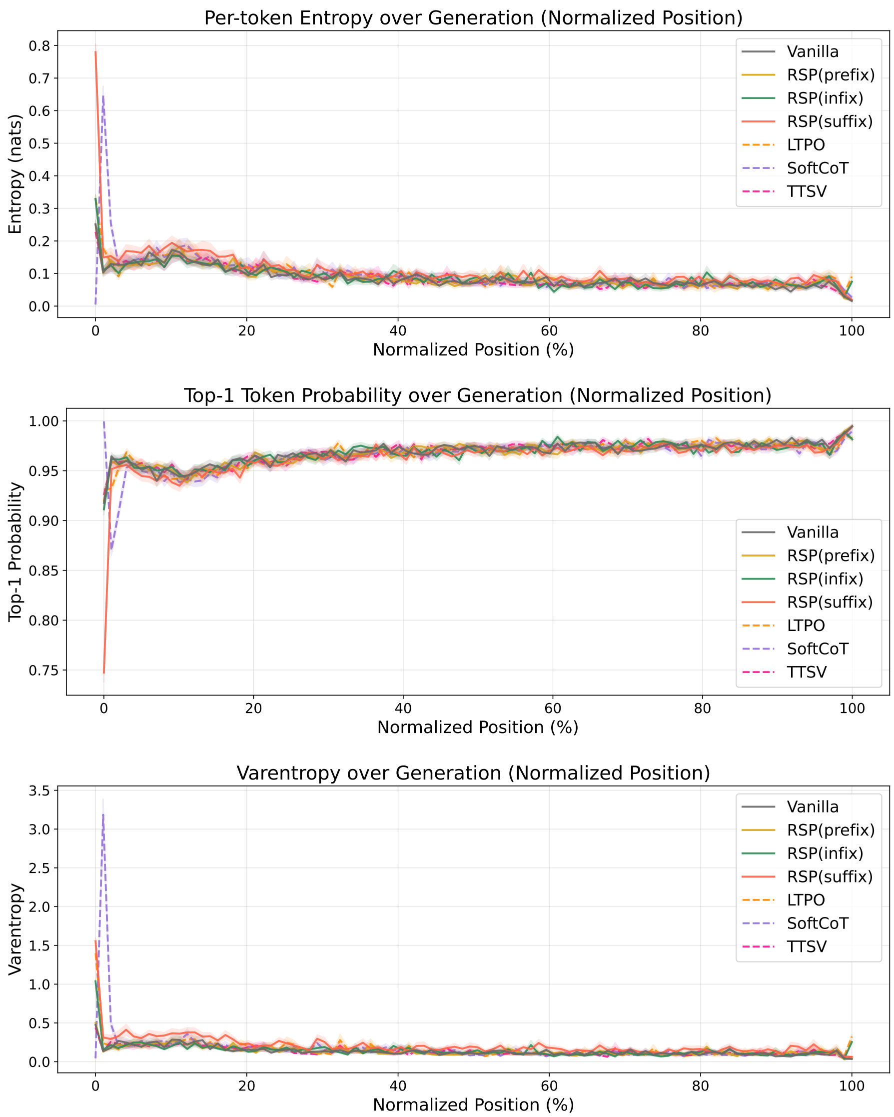


Figure 5: Mean entropy (top), top-1 probability (middle), and varentropy (bottom) over the full normalized generation trajectory (0–100%). Averaged over MATH-500, shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: existing soft prompt methods. All methods converge after the initial stage.

B Additional Semantic Diversity Analysis

This appendix complements the semantic diversity results reported in Section 3.4. We include per-problem t-SNE plots across a broader set of problems.

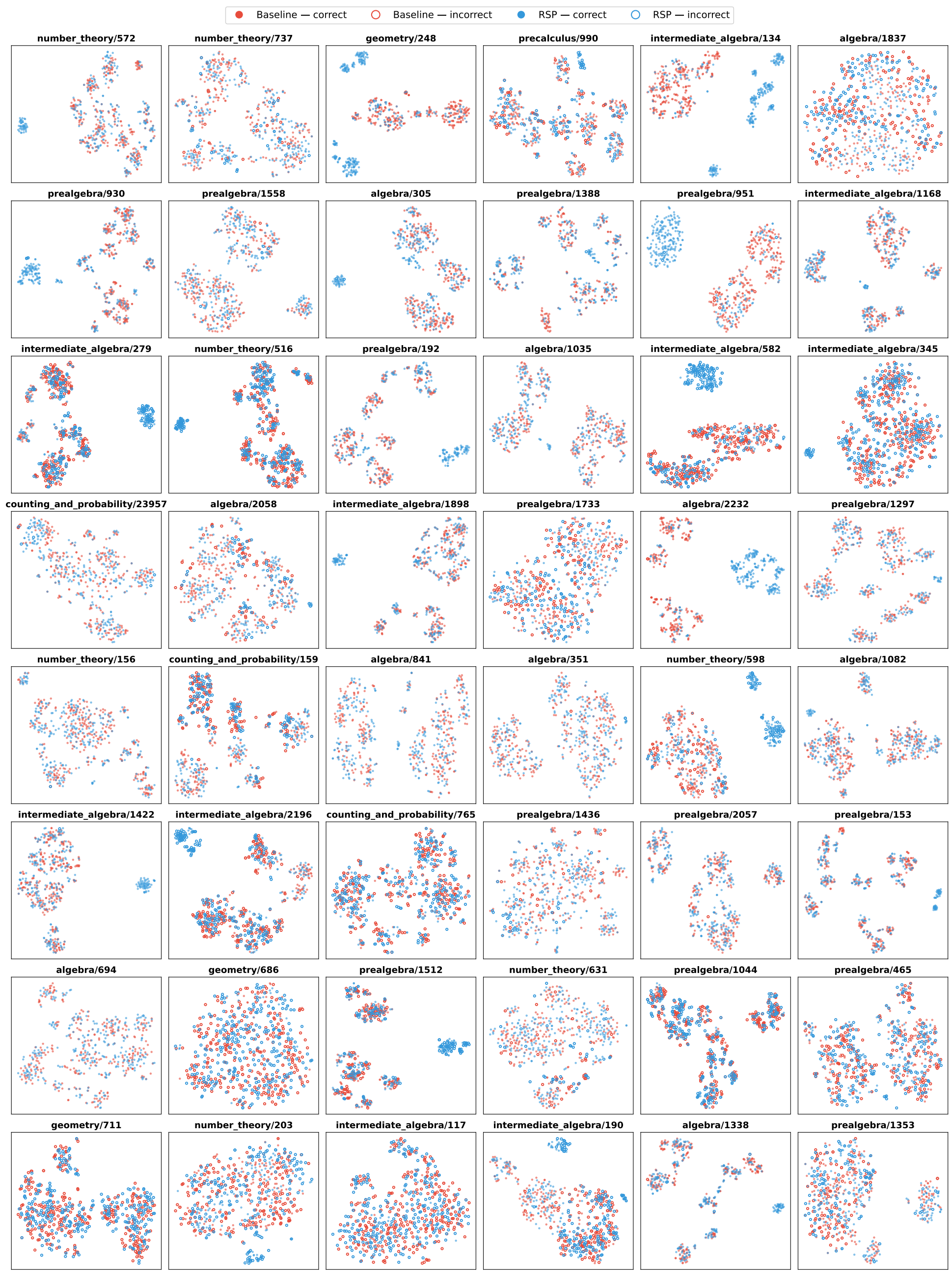


Figure 6: t-SNE projections of BGE-M3 embeddings for 48 out of the 64 evaluated MATH-500 problems (6×8 grid). Each subplot shows 300 baseline (red) and 300 RSP suffix (blue) samples under $\tau = 1.0$ with Qwen2.5-Math-1.5B-Instruct; filled markers denote correct answers and open markers denote incorrect. Subplot titles indicate `subject/problem_id`. Across most problems, RSP trajectories spread across multiple distinct regions while baseline samples concentrate in a single dominant cluster.

C Token-Level Distribution Metrics

This appendix provides formal definitions for the three metrics used in the Token-level analysis of Section 3.4. Let P_{base} denote the baseline next-token distribution at $\tau = 1.0$ and P_{target} the candidate distribution being compared (e.g., the baseline at $\tau = 2.0$ or RSP at $\tau = 1.0$). All metrics are computed analytically from saved logits at the first generation step.

Spearman rank correlation. Spearman's ρ is the Pearson correlation coefficient between the token ranks of two distributions:

$$\rho \;=\; \mathrm{Pearson}(\mathrm{rank}(P_{\text{target}}),\; \mathrm{rank}(P_{\text{base}}))\,. \tag{5}$$

Because temperature scaling $P^{(\tau)} \propto P^{1/\tau}$ is a strictly monotone transform, the rank order of tokens is preserved exactly, so $\rho = 1$ for any $\tau > 0$ as a mathematical identity. Any value $\rho < 1$ therefore indicates rank reorganization beyond what temperature scaling can produce.

Mass outside top-K. Let $\mathrm{TopK}(P_{\text{base}})$ be the set of K tokens with the highest probability under P_{base}. The probability mass that the candidate distribution places *outside* this set is

$$\mathrm{MassOutside}_K \;=\; 1 - \sum_{t \in \mathrm{TopK}(P_{\text{base}})} P_{\text{target}}(t). \tag{6}$$

This directly measures support expansion: how much probability the candidate assigns to tokens that are not in the baseline's preferred set. Reported values use $K = 10$.

Jensen–Shannon divergence. The symmetrized, bounded version of Kullback–Leibler divergence:

$$\mathrm{JS}(P, Q) \;=\; \tfrac{1}{2}\,\mathrm{KL}(P \,\|\, M) + \tfrac{1}{2}\,\mathrm{KL}(Q \,\|\, M), \qquad M = \tfrac{1}{2}(P + Q). \tag{7}$$

We use base-2 logarithms so that JS $\in [0, 1]$. Tokens absent from the stored top-100 are assigned a sentinel logit (-10^9) before softmax, which is negligible for our setting since the top-5 covers more than 99.9% of the probability mass.

D Proofs for Energy-Theoretic Analysis

D.1 Deterministic Trapping

Lemma 1 (Descent lemma)**.** *If E is L_E-smooth and $\eta \leq 1/L_E$, then $E(\boldsymbol{\sigma}^{\text{new}}) \leq E(\boldsymbol{\sigma}) - \frac{\eta}{2}\|\nabla E(\boldsymbol{\sigma})\|^2$.*

Proof. By L_E-smoothness, $E(\boldsymbol{\sigma} - \eta\nabla E) \leq E(\boldsymbol{\sigma}) - \eta\|\nabla E\|^2 + \frac{L_E\eta^2}{2}\|\nabla E\|^2 = E(\boldsymbol{\sigma}) - \eta(1 - \frac{L_E\eta}{2})\|\nabla E\|^2$. Since $\eta \leq 1/L_E$, we have $L_E\eta/2 \leq 1/2$, giving $E(\boldsymbol{\sigma}^{\text{new}}) \leq E(\boldsymbol{\sigma}) - \frac{\eta}{2}\|\nabla E\|^2$. $\square$

Theorem 1 (Deterministic trapping)**.** *Without perturbation, the forward pass converges to the local minimum $\boldsymbol{\sigma}^*$ in the basin of attraction of $\boldsymbol{\sigma}^{(0)}$, which may satisfy $E(\boldsymbol{\sigma}^*) > E(\boldsymbol{\sigma}_g^*)$.*

Proof. Lemma 1 guarantees monotone energy decrease at each layer: $E(\boldsymbol{\sigma}^{(l+1)}) \leq E(\boldsymbol{\sigma}^{(l)})$. Since E is bounded below (the quadratic term dominates), the sequence $\{E(\boldsymbol{\sigma}^{(l)})\}$ converges, implying $\|\nabla E(\boldsymbol{\sigma}^{(l)})\| \to 0$ and convergence to a critical point. Deterministic dynamics follow a unique trajectory from $\boldsymbol{\sigma}^{(0)}$ and cannot cross energy barriers to reach minima in other basins. $\square$

D.2 Properties of the RSP-Induced Perturbation

Decomposition of attention output. The attention update with n real keys and k random keys can be decomposed as

$$\mathbf{x}_i^{(l+1)} = (1 - w_{r,i}^{(l)})\,\tilde{\mathbf{x}}_i^{(l+1)} + \boldsymbol{\eta}_i^{(l)}, \tag{8}$$

where

$$\tilde{\mathbf{x}}_i^{(l+1)} = \sum_{j=1}^{n} \tilde{\alpha}_{ij}^{(l)} \mathbf{v}_j^{(l)}, \quad \tilde{\alpha}_{ij}^{(l)} = \frac{\alpha_{ij}^{(l)}}{1 - w_{r,i}^{(l)}}, \tag{9}$$

and

$$\boldsymbol{\eta}_i^{(l)} = \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)}. \tag{10}$$

We now analyze the properties of the perturbation term $\boldsymbol{\eta}_i^{(l)}$.

Conditional expectation. Conditioned on the attention weights, we have

$$\mathbb{E}\Big[\boldsymbol{\eta}_i^{(l)} \mid \alpha^{(l)}\Big] = \mu_E\, w_{r,i}^{(l)}, \tag{11}$$

where $w_{r,i}^{(l)} := \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)}$.

Conditional covariance. Assuming the random vectors are independent with covariance $\sigma_E^2\mathbf{I}$,

$$\operatorname{Cov}\Big(\boldsymbol{\eta}_i^{(l)} \mid \alpha^{(l)}\Big) = \sigma_E^2 \sum_{j=1}^{k} (\alpha_{i,n+j}^{(l)})^2\, \mathbf{I}. \tag{12}$$

Using $\sum_j \alpha_j^2 \leq (\sum_j \alpha_j)^2$, we obtain

$$\operatorname{Cov}\Big(\boldsymbol{\eta}_i^{(l)} \mid \alpha^{(l)}\Big) \preceq \sigma_E^2 (w_{r,i}^{(l)})^2\, \mathbf{I}. \tag{13}$$

Bias-to-variance ratio. The ratio between the mean and standard deviation is bounded by

$$\frac{\|\mathbb{E}[\boldsymbol{\eta}_i^{(l)} \mid \alpha^{(l)}]\|}{\sqrt{\operatorname{Var}(\boldsymbol{\eta}_i^{(l)} \mid \alpha^{(l)})}} \leq \frac{|\mu_E|}{\sigma_E}. \tag{14}$$

Empirically, $|\mu_E|/\sigma_E \ll 1$, implying that the stochastic component dominates the bias.

Table 9: Embedding matrix statistics. $|\mu_E|/\sigma_E < 0.02$ across all models.

Model	μ_E	σ_E	$\lvert\mu_E\rvert/\sigma_E$
LLaMA-3.1-8B-Instruct	0.00002	0.01057	0.002
Qwen2.5-Math-7B-Instruct	−0.00002	0.01647	0.001
Qwen2.5-Math-1.5B-Instruct	0.00019	0.01585	0.012
Qwen2.5-Math-7B	−0.00003	0.01842	0.001
Qwen2.5-Math-1.5B	0.00038	0.02990	0.013

Empirical verification. Table 9 reports the embedding matrix statistics for all models used in our experiments.

Implication. Therefore, the perturbation behaves as approximately zero-mean noise whose magnitude is controlled by $w_{r,i}^{(l)}$, justifying the Langevin-like interpretation in the main text.

D.3 Attention Decay Bound

Theorem 2 (Attention decay)**.** *The total attention on random tokens satisfies*

$$w_{r,i}^{(l)} \le \frac{k}{k + n \cdot \exp(\Delta_i^{(l)}/\sqrt{d})}, \tag{15}$$

where $\Delta_i^{(l)} := \min_{j\in[n]} \mathbf{q}_i^\top \mathbf{k}_j - \max_{j'\in[k]} \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}$ *is the real-vs-random logit gap.*

Proof. Let $s_j = \mathbf{q}_i^\top \mathbf{k}_j/\sqrt{d}$ for real keys and $s_{r_{j'}} = \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}/\sqrt{d}$ for random keys. By definition of $\Delta_i^{(l)}$, each real key satisfies $s_j \ge \Delta_i^{(l)}/\sqrt{d} + s_{r,\max}$ where $s_{r,\max} = \max_{j'} s_{r_{j'}}$. Therefore $\exp(s_j) \ge \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot \exp(s_{r,\max})$ for every real key. Summing over n real keys:

$$\sum_{j=1}^{n} \exp(s_j) \ge n \cdot \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot \exp(s_{r,\max}) \ge n \cdot \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot \frac{1}{k} \sum_{j'=1}^{k} \exp(s_{r_{j'}}). \tag{16}$$

The total attention on random tokens is $w_{r,i}^{(l)} = \sum_{j'} \exp(s_{r_{j'}})/(\sum_j \exp(s_j) + \sum_{j'} \exp(s_{r_{j'}}))$. Substituting the lower bound on the real-key sum and letting $R = \sum_{j'} \exp(s_{r_{j'}})$:

$$w_{r,i}^{(l)} = \frac{R}{\sum_j \exp(s_j) + R} \le \frac{R}{n \cdot \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot R/k + R} = \frac{k}{k + n \cdot \exp(\Delta_i^{(l)}/\sqrt{d})}. \tag{17}$$

□

D.4 Proofs for Output Distribution Shift

D.4.1 Autoregressive Formulation

In autoregressive decoding, the joint distribution over a sequence $y_{1:T}$ factorizes as

$$\Pr(y_{1:T} \mid x) = \prod_{t=1}^{T} p(y_t \mid x, y_{<t}),$$

where each step is governed by a prefix-conditioned distribution. Any perturbation introduced by RSP affects generation through a sequence of local changes to $p(y_t \mid x, y_{<t})$, rather than a single global distribution.

D.4.2 Local Linearization of Logits under RSP

Let h denote the random soft prompt. We model the perturbed logits as a function $z_i(h)$, assuming local smoothness in a neighborhood of $h = 0$:

$$z_i(h) = z_i(0) + \nabla_h z_i(0)^\top h + r_i(h),$$

where $r_i(h) = O(\|h\|^2)$. Defining $a_i := \nabla_h z_i(0)$, we obtain the local linear approximation $z_i(h) \approx z_i + a_i^\top h$. This approximation is valid for sufficiently small $\|h\|$ and is used to characterize the direction of perturbation rather than exact logit values.

D.4.3 Proof of Proposition 1 (Stochastic Ranking)

If there exists a token pair (i, j) such that $a_i \neq a_j$, then $0 < \Pr_h(z_i(h) > z_j(h)) < 1$.

Proof. Under the local linear approximation, $z_i(h) - z_j(h) = (z_i - z_j) + (a_i - a_j)^\top h$. Let $b_{ij} := a_i - a_j \neq 0$. If $h \sim \mathcal{N}(0, \sigma^2 I)$, then $b_{ij}^\top h \sim \mathcal{N}(0, \sigma^2 \|b_{ij}\|^2)$, a non-degenerate Gaussian. The perturbed logit difference $\Delta_{ij}(h) := (z_i - z_j) + b_{ij}^\top h$ therefore has non-zero probability mass on both sides of zero: $0 < \Pr_h(\Delta_{ij}(h) > 0) < 1$. □

D.4.4 On Proposition 2 (Support Expansion)

Proposition 2 states that tokens outside the baseline top-K set can enter the top-K set under RSP with positive probability. Proposition 1 establishes that pairwise ranking between tokens becomes stochastic. Entering the top-K set requires surpassing multiple tokens simultaneously, i.e., conditions of the form $z_i(h) > z_j(h)$ for all $j \in J$ where J is a subset of higher-ranked tokens. Under the local linear model, these inequalities define a region in h-space. If this region has non-zero measure under the distribution of h, then top-K entry occurs with positive probability. We interpret Proposition 2 as a support-expansion statement: RSP enables tokens that are effectively inaccessible under deterministic ranking to become reachable under stochastic perturbations.

D.4.5 Trajectory-Level Implications

Since autoregressive generation composes conditional distributions across steps, perturbations at early steps can lead to distinct trajectories. Let $\mathcal{R}_{k,K}(x)$ denote the set of trajectories containing at least one token outside the baseline top-K set. If RSP increases the accessibility of such tokens at early steps, then the probability of sampling trajectories in $\mathcal{R}_{k,K}(x)$ increases under repeated sampling. This links local ranking perturbations to global trajectory diversity.

E Attention Mass Analysis

This appendix formalizes the per-token attention mass reported in Figure 4 and the exclusion criteria applied to the reasoning span and the layer axis. For each of the 500 MATH-500 problems, a fresh RSP $\mathbf{H} \in \mathbb{R}^{L \times d}$ with $L = 10$ is sampled independently, so the reported heatmaps average over both problem variability and RSP-vector variability.

E.1 Per-Token Attention Mass

For each problem, we measure attention on the concatenated sequence [prompt; $\mathbf{H}$; generation], where the generation is the trajectory produced by the same model under matching RSP injection. Let $\alpha_{i,j}^{(\ell,h)}$ denote the post-softmax attention weight at layer ℓ, head h, from query position i to key position j, satisfying $\sum_j \alpha_{i,j}^{(\ell,h)} = 1$ under the causal mask. Let Q and R denote the index sets of question and RSP tokens with sizes $|Q|$ and $|R| = L$, and let H be the number of heads. The per-token attention mass that query i directs to each group at layer ℓ is

$$\mathrm{PTM}_Q[\ell, i] = \frac{1}{|Q|H} \sum_{h=1}^{H} \sum_{j \in Q} \alpha_{i,j}^{(\ell,h)}, \qquad \mathrm{PTM}_R[\ell, i] = \frac{1}{|R|H} \sum_{h=1}^{H} \sum_{j \in R} \alpha_{i,j}^{(\ell,h)}. \tag{18}$$

Normalization by $|Q|$ and $|R|$ controls for group size, so both quantities express how much attention a single question (resp. RSP) token receives on average under the same softmax denominator. Question tokens thereby serve as a size-matched baseline that absorbs sequence-length effects common to both groups.

E.2 Heatmap Aggregation

We partition the layer indices and the reasoning position indices each into five equal-size bins $\{B_L^{(b)}\}_{b=1}^5$ and $\{B_P^{(b)}\}_{b=1}^5$. For sample s and target group $\bullet \in \{Q, R\}$, the value in cell (b_L, b_P) is

$$M_\bullet^{(s)}[b_L, b_P] = \frac{1}{|B_L^{(b_L)}| \cdot |B_P^{(b_P)}|} \sum_{\ell \in B_L^{(b_L)}} \sum_{i \in B_P^{(b_P)}} \mathrm{PTM}_\bullet[\ell, i]. \tag{19}$$

Figure 4 reports the sample average of Eq. (19) over the 500 valid samples.

E.3 Excluding the `\boxed{...}` Span

The reasoning position range terminates strictly before the final `\boxed{...}` span. Chain-of-thought outputs decompose into a *text* (content) component and a *pattern* (template) component that contribute independently to downstream performance [Madaan and Yazdanbakhsh, 2022]. The `\boxed{...}` wrapper is a canonical pattern: a short, stereotyped span whose role is to complete the answer format rather than to extend reasoning. Including it would therefore inject a template-driven attention signature unrelated to reasoning dynamics.

E.4 Excluding the Final Two Layers

The layer axis further excludes the last two transformer layers. Two well-documented properties of late layers, none specific to our setup, motivate this choice.

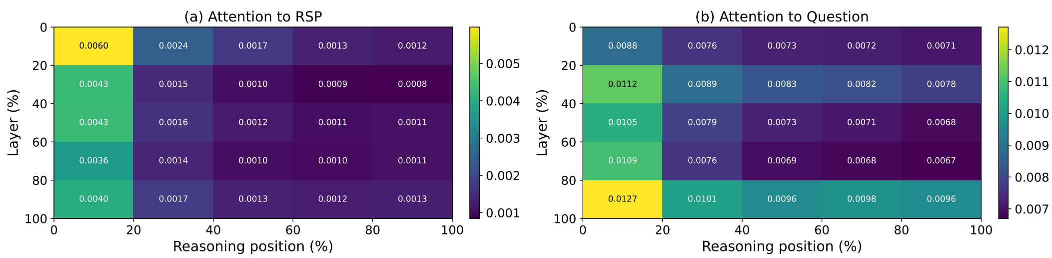
Output-projection specialization. Late-layer hidden states lie in an approximately linear relationship with vocabulary logits and therefore function as a progressive linear readout rather than as representation-transforming blocks [Belrose et al., 2023]. Consistently, late-layer feed-forward modules have been characterized as output-distribution shaping mechanisms [Geva et al., 2021]. Attention in these layers therefore reflects output formation rather than reasoning computation.

Register-token hypothesis. Uninformative patch tokens in Vision Transformers have been shown to be repurposed as *registers*: storage slots that absorb attention in deeper layers [Darcet et al., 2024]. Because RSPs share a similar profile as continuous tokens without pretrained semantic content, a similar repurposing in late transformer layers cannot be ruled out. Excluding the final two layers removes this potential register-induced confound from the reasoning-phase signal.

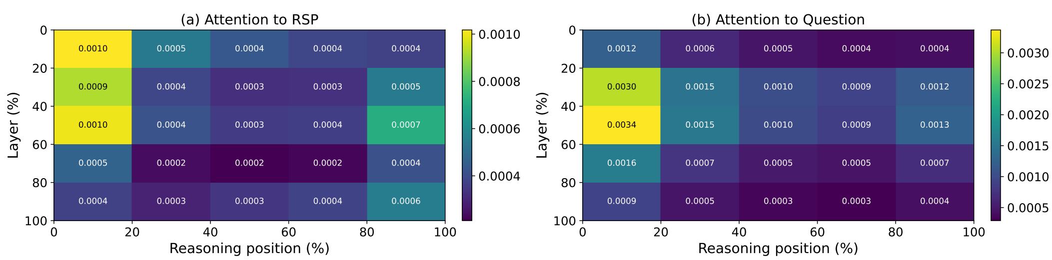
The two lines of evidence above identify known late-layer phenomena — linear readout and potential register repurposing — whose attention signatures are orthogonal to reasoning-phase dynamics. Removing the final two layers preempts their contamination of the reasoning-phase signal, yielding a principled, literature-grounded exclusion rather than a model-specific tuning decision.

E.5 Additional Suffix Heatmaps

Figure 7 reports the same per-token RSP attention analysis under suffix injection for Qwen2.5-Math-1.5B-Instruct and LLaMA-3.1-8B-Instruct. For the late-layer reasons discussed above, the pattern does not generalize uniformly across every cell; nevertheless, the overall trend is consistent across both models: question-token attention varies broadly across layers, whereas RSP attention shows the layer-wise and sequence-wise decay predicted by Eq. (4).



(a) Qwen2.5-Math-1.5B-Instruct, suffix



(b) LLaMA-3.1-8B-Instruct, suffix

Figure 7: Per-token RSP attention mass under suffix injection for the remaining two models (500 MATH-500 samples each). Axes and preprocessing match Figure 4: x-axis is reasoning position binned into five quantiles, y-axis is layer depth binned into five quantiles (top = shallow), color encodes the 500-sample mean per-token attention assigned to RSP tokens, and tokens inside `\boxed{}` spans and the last two layers are excluded.

F GRPO Formulation

For a given prompt q, the behavior policy $\pi_{\theta_{\text{old}}}$ samples a group of G responses $\{o_i\}_{i=1}^{G}$ with corresponding rewards $\{R_i\}_{i=1}^{G}$. The advantage of each response is computed by normalizing the group-level rewards:

$$\hat{A}_{i,t} = \frac{R_i - \text{mean}(\{R_i\}_{i=1}^{G})}{\text{std}(\{R_i\}_{i=1}^{G})}. \tag{20}$$

The policy is updated via a clipped objective with a KL penalty:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\Big(\min\Big(r_{i,t}\hat{A}_{i,t},\ \text{clip}(r_{i,t}, 1-\varepsilon, 1+\varepsilon)\,\hat{A}_{i,t}\Big) - \beta D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})\Big)\right], \tag{21}$$

where $r_{i,t} = \pi_\theta(o_{i,t} \mid q, o_{i,<t})/\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})$ is the importance sampling ratio, ε is the clipping range, and β controls the strength of the KL penalty against the reference policy π_{ref}.

G Broader Impacts

This work provides empirical and theoretical analysis of how random embedding injection affects the reasoning behavior of large language models. The contributions are primarily foundational, with potential downstream implications for reinforcement learning-based post-training of reasoning models such as GRPO.

Positive impacts. RSP-induced trajectory diversity could improve the sample efficiency of RL-based LLM post-training by providing richer learning signals within group-relative advantage estimation. This has the potential to reduce compute costs associated with alignment and reasoning-oriented fine-tuning, making such methods more accessible to research groups with limited resources.

Potential concerns. Methods that expand the effective output support of LLMs may increase the reachability of low-probability tokens, which includes both desirable alternative reasoning paths and potentially unsafe or off-distribution outputs. Practitioners applying RSP in production systems should combine it with existing safety filtering and alignment mechanisms rather than treating it as a standalone diversity source.

Limitations on impact scope. Since this work focuses on analysis rather than deploying a new model or dataset, the direct societal impact is limited. The broader implications described above are contingent on future work integrating RSP into applied training pipelines.